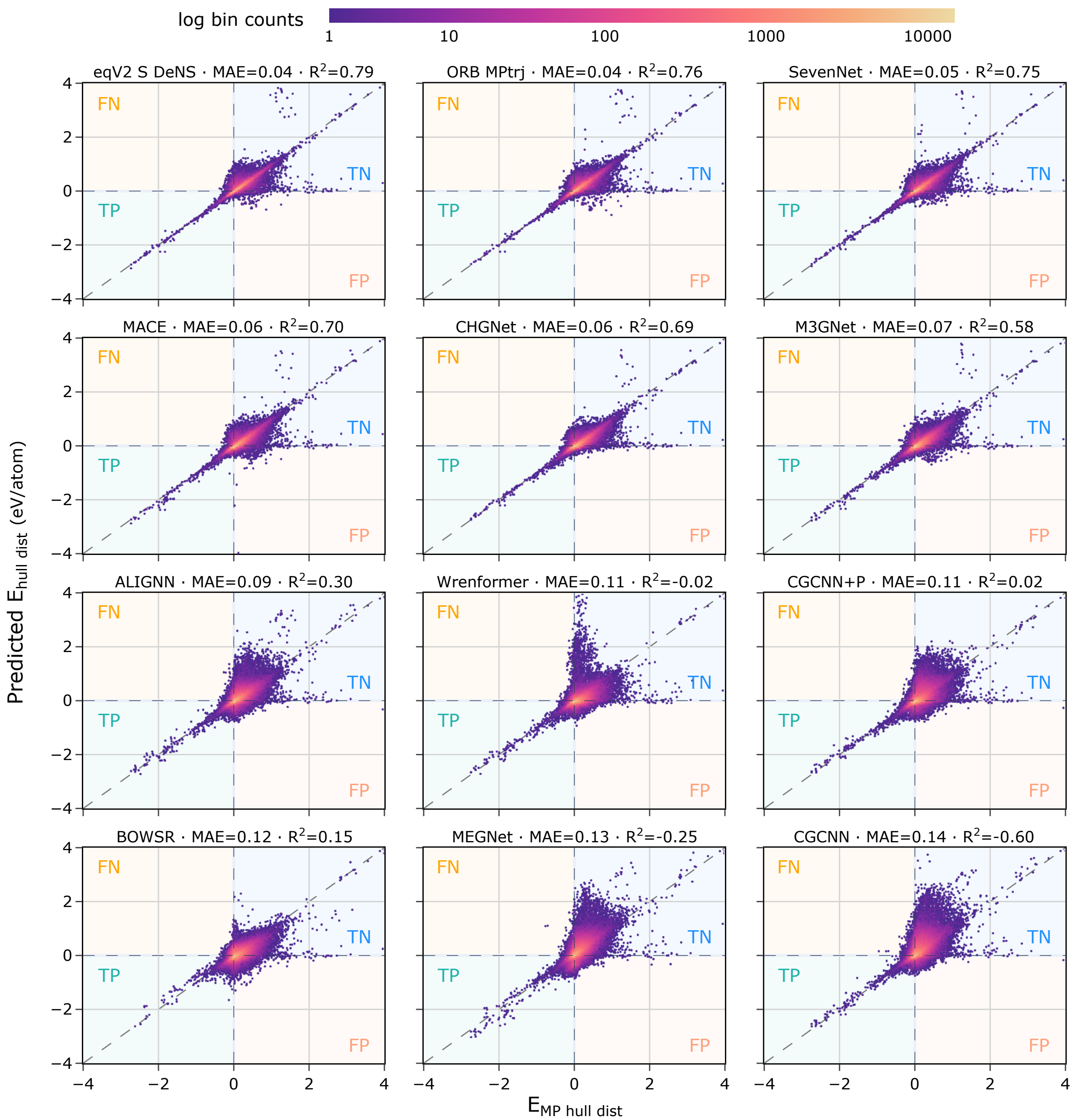

Figure S2: Parity plots of model-predicted energy distance to the convex hull (based on their formation energy predictions) vs DFT ground truth, color-coded by log density of points. Models are sorted left to right and top to bottom by MAE. For parity plots of formation energy predictions, see fig. S3.

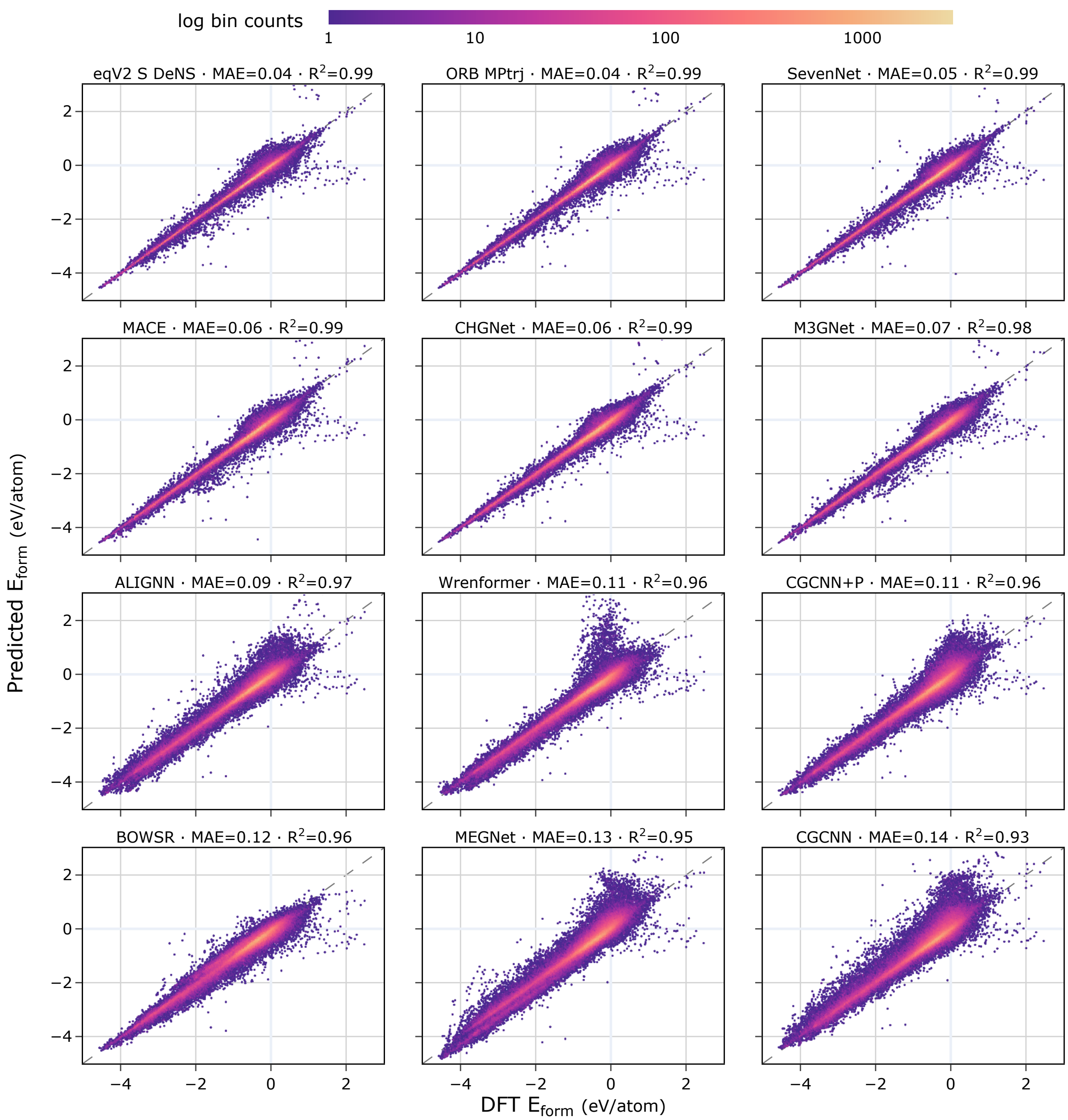

Figure S3: Parity plots of model-predicted formation energies vs DFT formation energies, color-coded by log density of points. The models are sorted from left to right and top to bottom by MAE. While similar to the parity plots in fig. S2 which shows the predicted distance to the convex hull vs DFT ground truth, this figure better visualizes the point density due to formation energy's wider spread. We observe broadly the same failure modes with occasional high DFT energy outliers predicted as near 0 formation energy by the models.

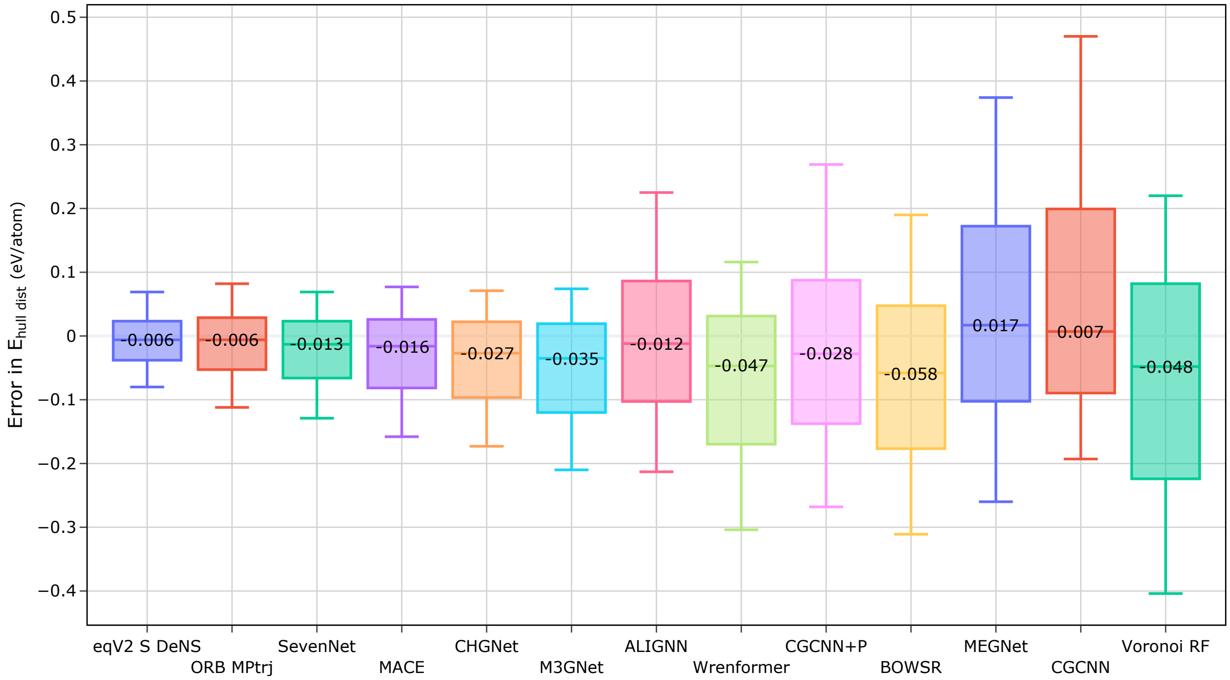

Figure S4: Box plot of interquartile ranges (IQR) of hull distance errors for each model. The whiskers extend to the 5th and 95th percentiles. The horizontal line inside the box shows the median. BOWSR has the highest median error, while Voronoi RF has the highest IQR.

formance decrease for Wrenformer and Voronoi RF than for UIPs and even force-less GNNs with larger errors like ALIGNN, MEGNet and CGCNN.

Figure S6 shows the rolling MAE for different models split out by calculation batch in the WBM data set. MEGNet and CGCNN both incur a higher rolling MAE than Wrenformer across all 5 batches and across most or all of the hull distance range visible in these plots. However, similar to the UIPs, MEGNet and CGCNN exhibit very little degradation for higher batch counts. The fact that higher errors for later batches are specific to Wrenformer and Voronoi RF suggests that training only on composition and coarse-grained structural features (spacegroup and Wyckoff positions in the case of Wrenformer; coordination numbers, local environment properties, etc. in the case of Voronoi RF) alone is insufficient to learn an extrapolatable map of the PES.

Given its strong performance on batch 1, it is possible that given sufficiently diverse training data, Wrenformer could become similarly accurate to the UIPs across the whole PES landscape at substantially less training and inference cost. However, the loss of predictive forces and stresses may make Wrenformer unattractive for certain applications even then.

Note S1.7 Largest Errors vs. DFT Hull Distance

Given the large variety of models tested, we asked whether any additional insight into the errors can be gained from looking at how the predictions vary between different models. In fig. S7 we see two distinct groupings emerge when looking at the 200 structures with the largest errors. This clustering is particularly apparent when points are colored by model disagreement.

Note S1.8 Exploratory Data Analysis

To give high-level insights into the MP training and WBM test set used in this work, we include element distributions for structures in both datasets (fig. S8). To show how frame selection from MP structure relaxation affected relative elemental abundance between MP relaxed structures and the snapshots in MPtrj, fig. S9 shows element occurrence ratios between MPtrj and MP. Similarly, fig. S10 shows the elements-per-structure distribution of MP, MPtrj and WBM, normalized by dataset size. The mode of all three datasets is 3, but WBM's share of ternary phases is noticeably more peaked than MP's, which includes small numbers of unary and senary phases. fig. S11 plots the distributions of formation energies, decomposition energies and convex hull distances (with respect to the convex hull spanned by MP materials only) for the MP and WBM data sets. This plot not only gives insight into the nature of the dataset but also emphasizes the increased difficulty of stability vs formation energy prediction arising from the much narrower distribution of convex hull distances compared to the more spread-out formation energy distribution. For WBM we see that the formation energy distribution exhibits much wider spread than the convex hull distance distribution, spanning al-

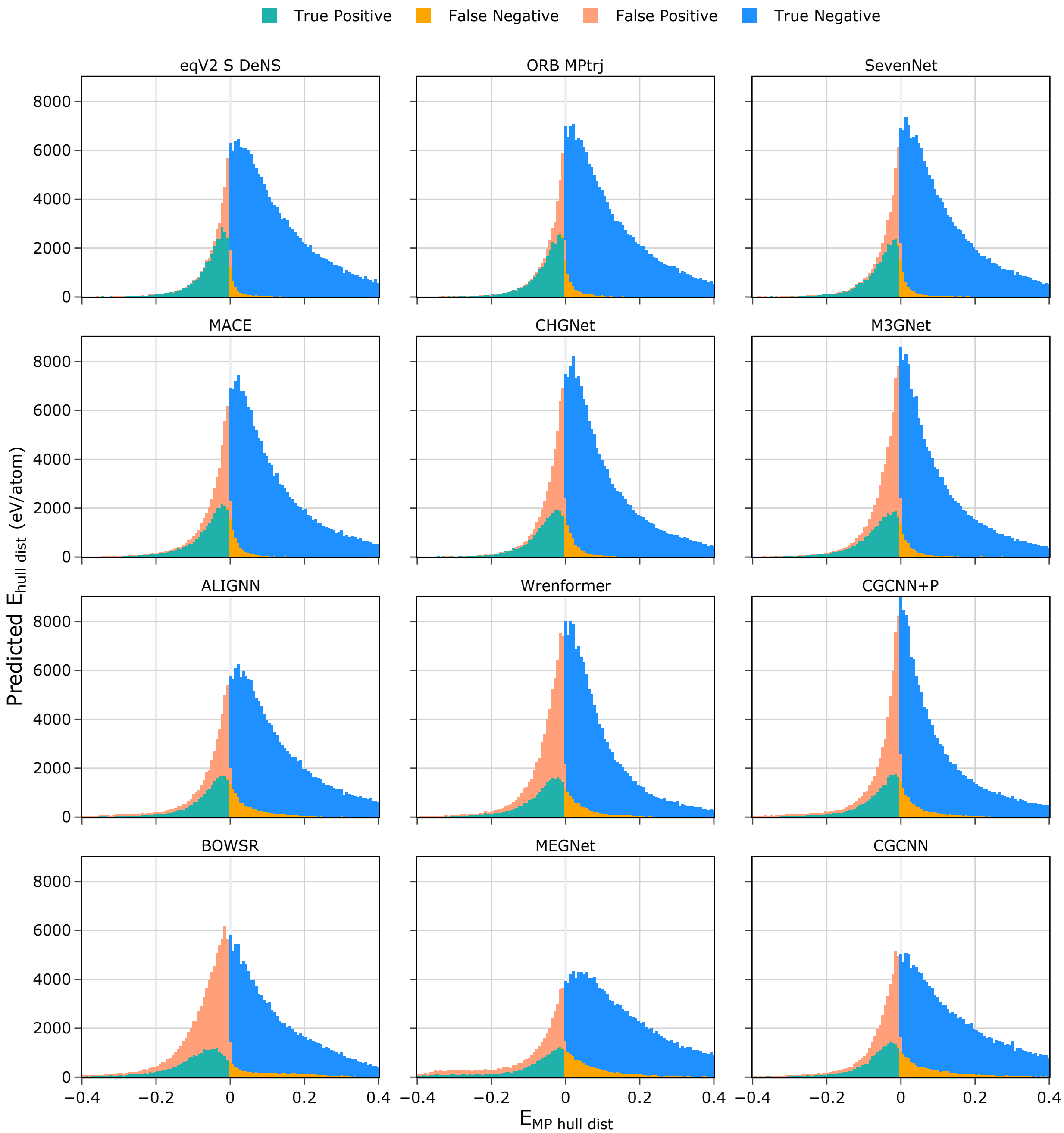

Figure S5: Distribution of model-predicted hull distance colored by stability classification. Models are sorted from top to bottom by F1 score. The thickness of the red and yellow bands shows how often models misclassify as a function of how far away from the convex hull they place a material.

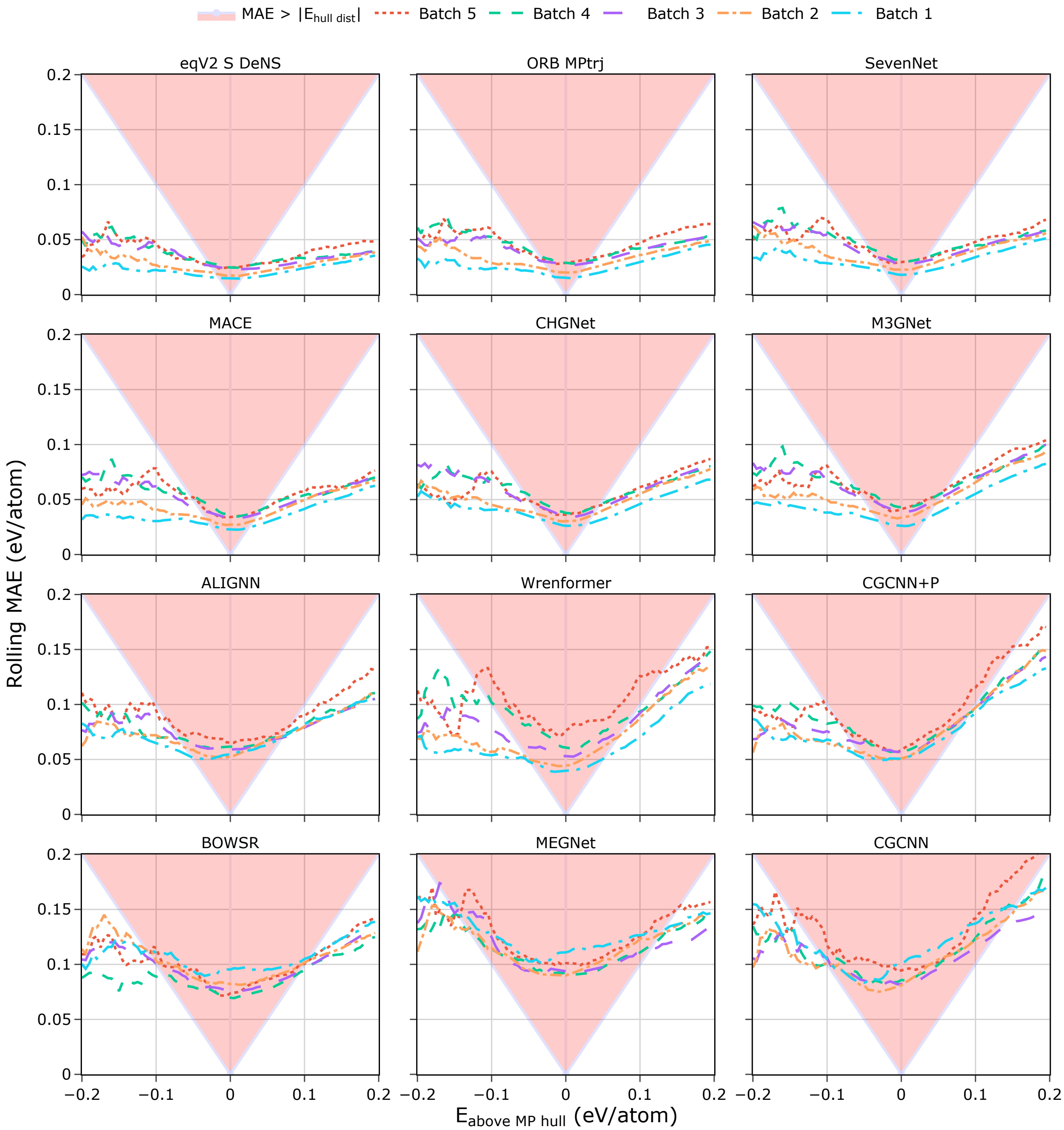

Figure S6: Rolling MAE as a function of distance to the convex hull for different models. Most models considered show a predictable decrease in performance from batch 1 to batch 5. The effect is larger for some models than others but batches 4 and 5 consistently incur the highest convex hull distance errors for all models. The UIPs show stronger extrapolative performance, as they show minimal deterioration in performance on later batches that move further away from the original MP training distribution. Wrenformer, by contrast, exhibits a pronounced increase in MAE with batch count. We view these plots as a strong indicator that Matbench Discovery is indeed testing out-of-distribution extrapolation performance as it is Occam's razor explanation for the observed model performance drop with increasing batch count.